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ECOSYSTEM Technical Group 4

Overview of standardisation Committees, published and under development standards, and gap analysis for renewable materials/ textiles

Lead authors: K Eufinger (Centexbel), H Vilaça (Citeve)

Additional contributions: C. Marchall (RISE), STAR4BBS, ECOSYSTEM TG4 members

Executive Summary

Standards provide a framework for ensuring consistency, reliability, and safety in various aspects of life, from technology and manufacturing to health care and agriculture. By following standards, organizations and individuals can ensure that their products, processes, and services meet the necessary requirements and are compatible with those of others, promoting interoperability, innovation, and market growth. Next to standards there are also private certification schemes and labels for bio-based and renewably sourced textiles and textile materials. Despite the huge effort being made on the development of new standards, these do not yet cover all materials and properties.

The first section of this document provides an inventory of the existing (published standards) and under development (work program) standards directly or indirectly related with renewable / bio-based materials and recycled ones, with a focus on textile materials, as that is the scope of this work. The second section of the document provides a gap analysis indicating future needs for standards development. In the third section a non-exhaustive overview over the existing private labels and certification schemes is provided.

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Overview of standardisation committees, published and under development standards

Standards are agreed-upon ways of doing something, typically established by experts in a subject matter to ensure consistency, reliability, and safety in various activities and industries. They can be mandatory or voluntary, and they provide a common language, guidelines, and technical requirements for diverse fields. Standards are often developed through collaboration between various stakeholders, such as manufacturers, sellers, buyers, customers, trade associations, and regulators.

They establish accepted practices and terminologies, ensuring that materials, products, processes, and services are fit for their purpose. Standards can help drive innovation, increase productivity, and protect consumers by ensuring safety, durability, and market equity.

Standards can be developed by various organizations, including national standardization bodies and international organizations such as the International Organization for Standardization (ISO),¹ the European Committee for Standardization (CEN),² the American Association of Textile Chemists and Colorists (AATCC),³ or ASTM International.⁴ These organizations facilitate the creation and adoption of standards, ensuring that they are widely accepted and applicable across different countries and industries.

¹ <https://www.iso.org/home.html>

² <https://www.cencenelec.eu/european-standardization/cen-and-cenelec/>

³ <https://www.aatcc.org/>

⁴ <https://www.astm.org/>

In essence, standards provide a framework for ensuring consistency, reliability, and safety in various aspects of life, from technology and manufacturing to health care and agriculture. By following standards, organizations and individuals can ensure that their products, processes, and services meet the necessary requirements and are compatible with those of others, promoting interoperability, innovation, and market growth.

Standards are typically updated and revised every three to five years, with revision cycles that can begin twice each year. This periodic review and update process ensures that standards remain current, relevant, and aligned with the latest industry practices and technological advancements.

Below is an inventory of the existing (published standards) and under development (work program) standards **directly or indirectly related with renewable / bio-based materials and recycled ones**, with a focus on **textile materials**, as that is the scope of this work. Also, the inventory is limited to standardization on the European Union.

The inventory is divided by the following Technical Committees, identified as working on relevant areas:

- CEN/TC 411 BIO-BASED PRODUCTS
- ISO/TC 38 TEXTILES
- ISO/TC 38 TEXTILES - SC 23 - FIBRES AND YARNS
- CEN/TC 248 TEXTILES AND TEXTILE PRODUCTS
- CEN/TC 134 RESILIENT, TEXTILE, LAMINATE AND MODULAR MECHANICAL LOCKED FLOOR COVERINGS
- CEN/TC 189 GEOSYNTHETICS
- ISO TC61 PLASTICS – SC14 ENVIRONMENTAL ASPECTS
- CEN/TC 249 PLASTICS
- CEN/TC 261 PACKAGING – SC 4 PACKAGING AND ENVIRONMENT
- CEN/TC 466 CIRCULARITY AND RECYCLABILITY OF FISHING GEAR AND AQUACULTURE EQUIPMENT

Remark: This inventory reflects the status at the moment of writing this document. Since standards development is continuously evolving the list will become outdated with time. For the up to date information the reader is referred to the respective website of the committee, which is provided in the overview below.

CEN TC411 Bio-based products ⁵

- Development of standards for bio-based products covering horizontal aspects. This includes consistent terminology, sampling, certification tools, bio-based content, application of and correlation towards life cycle analysis, sustainability criteria for biomass used and for final products, and aspects where further harmonization is needed on horizontal level;

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:874780&cs=18C173D69008BCCEB753FFD8DCAA03196

- Development of standards for bio-solvents, covering product functionality, biodegradability and, if necessary, product specific aspects not covered under i.

This TC has currently four Working Groups (WGs), 15 published standards and 3 standards under development.

Working group	Published standards	Work programme
CEN/TC 411/WG 1 "Bio-based products – Terminology" ⁶	• EN 16575:2014 Bio-based products - Vocabulary	Currently none
CEN/TC 411/WG 3 "Bio-based products – Bio-based content" ⁷	• CEN/TR 16721:2014 Bio-based products - Overview of methods to determine the bio-based content • EN 16640:2017 Bio-based products - Bio-based carbon content - Determination of the bio-based carbon content using the radiocarbon method • EN 16640:2017/AC:2017 Bio-based products - Bio-based carbon content - Determination of the bio-based carbon content using the radiocarbon method • EN 16785-1:2015 Bio-based products - Bio-based content - Part 1: Determination of the bio-based content using the radiocarbon analysis and elemental analysis • EN 16785-2:2018 Bio-based products - Bio-based content - Part 2: Determination of the bio-based content using the material balance method • EN 17351:2020 Bio-based products - Determination of the oxygen content using an elemental analyser	• prEN 16785-1 rev Bio-based products - Bio-based content - Part 1: Determination of the bio-based content using the radiocarbon analysis and elemental analysis (preliminary) • prEN 16640 rev (WI=00411020) Bio-based products - Bio-based carbon content - Determination of the bio-based carbon content using the radiocarbon method (active) • prEN 16640 rev Bio-based products - Bio-based carbon content - Determination of the bio-based carbon content using the radiocarbon method (active)
CEN/TC 411/WG 4 "Bio-based products - Sustainability criteria, life cycle analysis and related issues" ⁸	• CEN/TR 16957:2016 Bio-based products - Guidelines for Life Cycle Inventory (LCI) for the End-of-life phase • CEN/TR 17341:2019 Bio-based products - Examples of	• FprEN 18027 "Biobased Products – Life Cycle Assessment Additional requirements and guidelines for comparing the life cycles of biobased products with

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:904040&cs=1C4F24616EF7C4AB05C76B21A61335F56

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:904047&cs=1FFD5A7A7D3914E7857AEF064035657AB

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:904048&cs=113BC7C93BCD9AD3919606D549666DB35

Working group	Published standards	Work programme
	- reporting on sustainability criteria • EN 16751:2016 Bio-based products - Sustainability criteria • EN 16760:2015 Bio-based products - Life Cycle Assessment	their fossilbased equivalents" (under Approval)
CEN/TC 411/WG 5 " Bio-based products - Certification and declaration tools" (no active projects) ⁹	• EN 16848:2016 Bio-based products - Requirements for Business to Business communication of characteristics using a Data Sheet • EN 16935:2017 Bio-based products - Requirements for Business-to-Consumer communication and claims	Currently none

ISO/TC38 Textiles ¹⁰

Standardization of:

1. fibres, yarns, threads, cords, rope, cloth and other fabricated textile materials; and the methods of test, terminology and definitions relating thereto;
2. textile industry raw materials, auxiliaries and chemical products required for processing and testing;
3. specifications for textile products;
4. micro plastics from textile source and the methods of test, specifications, terminology and definitions relating thereto;
5. traceability and responsible sourcing of animal fibres in the textile supply chain and the methods of test, specifications, terminology and definitions relating thereto;
6. ethical and environmental issues in the textile supply chain and the methods of test, specifications, terminology, and definitions relating thereto.

This TC has currently fourteen WGs, 177 published standards and 23 standards under development.

Working group	Published standards	Work programme
ISO/TC 38/WG 30 "Textiles – Tests for biodegradability"	• ISO 21701:2019 Textiles —Test method for accelerated hydrolysis of textile materials and biodegradation under controlled composting conditions of the resulting hydrolysate • ISO 11721-1:2001 Textiles — Determination of resistance of cellulose-containing textiles to micro-organisms — Soil burial test — Part 1:	• ISO/DIS 21701(rev) Textiles — Test method for accelerated hydrolysis of textile materials and biodegradation under controlled composting conditions of the resulting hydrolysate • ISO/WD 24304 Textiles — Determination of the aerobic biodegradation of fibres and textile materials in seawater by measuring the

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https://standards.cencenelec.eu/dyn/www/?p=205:7:0:::FSP_ORG_ID:904049&cs=1BED99CFAF0A979A5A2D0D30398EC5380

¹⁰ <https://www.iso.org/committee/48148.html>

Working group	Published standards	Work programme
	Assessment of rot-retardant finishing (joint with CEN) ISO 11721-2:2003 Textiles — Determination of the resistance of cellulose-containing textiles to micro-organisms — Soil burial test — Part 2: Identification of long-term resistance of a rot retardant finish (joint with CEN)	biochemical oxygen demand or the amount of carbon dioxide evolved. ISO/NP 17952 Textiles — Test method for determination of degradation rate of textile materials under simulated composting conditions in a laboratory-scale test
ISO/TC 38/WG 31 “Textiles – Non-fibrous bio-based material for textiles”	- ISO 20921:2019 Textiles — Determination of stable nitrogen isotope ratio in cotton fibres - ISO 22195-1:2023 Textiles — Determination of index ingredient from coloured textile — Part 1: Madder - ISO 22195-2:2023 Textiles — Determination of index ingredient from coloured textile — Part 2: Turmeric - ISO 22195-3:2023 Textiles — Determination of index ingredient from coloured textile — Part 3: Myrobalan - ISO 22195-4:2021 Textiles — Determination of index ingredient from coloured textile — Part 4: Catechu - ISO 22195-5:2021 Textiles — Determination of index ingredient from coloured textile — Part 5: Lac - ISO 22195-6:2021 Textiles — Determination of index ingredient from coloured textile — Part 6: Punica granatum - ISO 22195-7:2024 Textiles — Determination of index ingredient from coloured textile — Part 7: Himalayan rhubarb (under publication) - ISO 22195-8:2024 Textiles — Determination of index ingredient from coloured textile — Part 8: Hibiscus (under publication)	ISO/CD 22195-9 Textiles — Determination of index ingredient from coloured textile — Part 9: Gallnut
ISO/TC 38/WG 33 “Animal welfare in the textile supply chain”	ISO 4465:2022 Textiles — Animal welfare in the supply chain — General requirements for the production, preparation and traceability of Angora rabbit fibre, including ethical claims and supporting information	ISO/WD 22786 Textiles — Animal welfare in the supply chain — General requirements for the production, preparation and traceability of cashmere fibre, including ethical claims and supporting information

Working group	Published standards	Work programme
ISO/TC 38/WG 34 "Textiles – Microplastics from textile sources"	- ISO 4484-1:2023 Textiles and textile products — Microplastics from textile sources — Part 1: Determination of material loss from fabrics during washing - ISO 4484-2:2023 Textiles and textile products — Microplastics from textile sources — Part 2: Qualitative and quantitative analysis of microplastics - ISO 4484-3:2023 Textiles and textile products — Microplastics from textile sources — Part 3: Measurement of collected material mass released from textile end products by domestic washing method	ISO/NP 4484-4 Textiles and textile products — Microplastics from textile sources — Part 4: Quantitative thermal or gravimetric analysis of certain synthetic materials released from fabrics during washing (under ballot for starting work)
ISO/TC 38/WG 35 "Textiles – Environmental aspects"	ISO 5157:2023 Textiles – Environmental aspects – Vocabulary	Currently none. Waiting for CEN/TC248/WG39 to advance with their work.

ISO/TC38 Textiles - SC23 - Fibres and Yarns ¹¹

Standardization of terminology, test methods, specifications and quality labelling of fibres and yarns.

This SC has currently three WGs, 55 published standards and 4 standards under development.

Working group	Published standards (selection of relevant documents, please refer to the website for the full list)	Work programme
ISO/TC 38/SC 23/WG 2 "Textiles - Fibres and yarns - Fibres - Natural cellulosic"	- ISO 12834:2024 Textiles — Synthetic filament yarns — Determination of dynamic thermal draw-force of partially oriented yarns (POY) - ISO 137:2015 Wool — Determination of fibre diameter — Projection microscope method - ISO 920:1976 Wool — Determination of fibre length (barbe and hauteur) using a comb sorter	ISO/CD 1130 Textile fibres — Some methods of sampling for testing
ISO/TC 38/SC 23/WG 5 "Textiles - Fibres and yarns -Fibres - Natural proteins"	- ISO 1130:1975 Textile fibres — Some methods of sampling for testing - ISO 1136:2015 Wool — Determination of mean diameter of fibres — Air permeability method	ISO/CD 1139 Textiles — Designation of yarns
ISO/TC 38/SC 23/WG 6 "Textiles - Fibres and yarns -Fibres - Man-made"	- ISO 1139:1973 Textiles — Designation of yarns - ISO 2061:2015 Textiles — Determination of twist in yarns — Direct counting method - ISO 2370:2019 Textiles — Determination of fineness of flax fibres — Permeametric methods	ISO/AWI 8159 Textiles — Morphology of fibres and yarns — Vocabulary

¹¹ <https://www.iso.org/committee/48334.html>

Working group	Published standards (selection of relevant documents, please refer to the website for the full list)	Work programme
	• ISO 2403:2021 Textiles — Cotton fibres — Determination of micronaire value • ISO 2646:1974 Wool — Measurement of the length of fibres processed on the worsted system, using a fibre diagram machine • ISO 2647:2020 Wool — Determination of percentage of medullated fibres by the projection microscope • ISO 2648:2020 Wool — Determination of fibre length distribution parameters — Capacitance method • ISO 2913:1975 Wool — Colorimetric determination of cystine plus cysteine in hydrolysates • ISO 2915:1975 Wool — Determination of cysteic acid content of wool hydrolysates by paper electrophoresis and colorimetry • ISO 2916:1975 Wool — Determination of alkali content • ISO 3060:1974 Textiles — Cotton fibres — Determination of breaking tenacity of flat bundles • ISO 3072:1975 Wool — Determination of solubility in alkali • ISO 3073:1975 Wool — Determination of acid content • ISO 3074:2014 Wool — Determination of dichloromethane-soluble matter in combed sliver • ISO 4912:1981 Textiles — Cotton fibres — Evaluation of maturity — Microscopic method • ISO 4913:1981 Textiles — Cotton fibres — Determination of length (span length) and uniformity index • ISO 5079:2020 Textile fibres — Determination of breaking force and elongation at break of individual fibres • ISO 5162:2023 Textiles — Quality labelling specification for dehaired cashmere • ISO 6939:1988 Textiles — Yarns from packages — Method of test for breaking strength of yarn by the skein method • ISO 10290:2018 Textiles — Cotton yarns — Basis for specification • ISO 10306:2014 Textiles — Cotton fibres — Evaluation of maturity by the air flow method • ISO 12027:2012 Textiles — Cotton-fibre stickiness — Detection of sugar by colour reaction • ISO 15625:2014 Silk — Electronic test method for defects and evenness of raw silk • ISO 17202:2002 Textiles — Determination of twist in single spun yarns — Untwist/retwist method	

Working group	Published standards (selection of relevant documents, please refer to the website for the full list)	Work programme
	- ISO 18066:2015 Textiles — Manmade filament yarns — Determination of shrinkage in boiling water - ISO 18068:2014 Cotton fibres — Test method for sugar content — Spectrophotometry - ISO 18596:2015 Test method for staple length of dehaired cashmere — Hand-arranging method - ISO 20754:2018 Textiles — Man-made fibres — Determination of shape factors in cross section - ISO 21046:2018 Silk — Test method for determining the size of silk yarns	

CEN/TC248 Textiles and Textile Products¹²

Standardization of the following aspects of textiles, textile products and textile components of products: 1) test methods; 2) terms and definitions; 3) specifications, and if necessary classifications, in terms of their expected behaviour, in particular where required by other CEN Technical Committees or the CEC or EFTA. Equipment relevant for the testing and use of textiles.

This TC has currently seventeen WGs, a few hundred published standards and 34 standards under development.

Working group	Published standards	Work programme
	EN ISO 6938:2014 Textiles - Natural fibres - Generic names and definitions (ISO 6938:2012)	
CEN/TC 248/WG 30 “Textiles and textile products - Composition - Qualitative and quantitative analysis of fibres and fibre mixtures” ¹³	- EN ISO 17751-1:2023 Textiles - Quantitative analysis of cashmere, wool, other specialty animal fibers and their blends - Part 1: Light microscopy method (ISO 17751-1:2023) - EN ISO 17751-2:2023 Textiles - Quantitative analysis of cashmere, wool, other specialty animal fibres and their blends - Part 2: Scanning electron microscopy method (ISO 17751-2:2023) - EN ISO 1833-14:2019 Textiles - Quantitative chemical	- prEN ISO 2076 rev Textiles — Man-made fibres — Generic names - prEN ISO 1833-1 rev Textiles — Quantitative chemical analysis — Part 1: General principles of testing

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https://standards.cencenelec.eu/dyn/www/?p=205:7:0:::FSP_ORG_ID:6229&cs=10D5E9507DBA0195D1D470609DD93A3DC

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https://standards.cencenelec.eu/dyn/www/?p=205:7:0:::FSP_ORG_ID:627430&cs=11BB54CAAADF2A474A0C2526995CE8A52

Working group	Published standards	Work programme
	analysis - Part 14: Mixtures of acetate with certain other fibres (method using glacial acetic acid) (ISO 1833-14:2019) • EN ISO 1833-15:2019 Textiles - Quantitative chemical analysis - Part 15: Mixtures of jute with certain animal fibres (method by determining nitrogen content) (ISO 1833-15:2019) • EN ISO 1833-17:2020 Textiles - Quantitative chemical analysis - Part 17: Mixtures of cellulose fibres and certain fibres with chlorofibres and certain other fibres (method using concentrated sulfuric acid) (ISO 1833-17:2019) • EN ISO 1833-18:2020 Textiles - Quantitative chemical analysis - Part 18: Mixtures of silk with wool or other animal hair (method using sulfuric acid) (ISO 1833-18:2020) • EN ISO 1833-1:2020 Textiles - Quantitative chemical analysis - Part 1: General principles of testing (ISO 1833-1:2020) • EN ISO 1833-22:2021 Textiles - Quantitative chemical analysis - Part 22: Mixtures of viscose or certain types of cupro or modal or lyocell with flax fibres (method using formic acid and zinc chloride) (ISO 1833 22:2020) • EN ISO 1833-3:2020 Textiles - Quantitative chemical analysis - Part 3: Mixtures of acetate with certain other fibres (method using acetone) (ISO 1833-3:2020) • EN ISO 1833-4:2023 Textiles - Quantitative chemical analysis - Part 4: Mixtures of certain protein fibres with certain other fibres (method using hypochlorite) (ISO 1833-4:2023) • EN ISO 1833-9:2019 Textiles - Quantitative chemical analysis - Part 9: Mixtures of acetate with certain other fibres (method using benzyl alcohol) (ISO 1833-9:2019) • EN ISO 20418-1:2018 Textiles - Qualitative and quantitative proteomic analysis of some	

Working group	Published standards	Work programme
	animal hair fibres - Part 1: Peptide detection using LC-ESI-MS with protein reduction (ISO 20418-1:2018) • EN ISO 20418-3:2020 Textiles - Qualitative and quantitative proteomic analysis of some animal hair fibres - Part 3: Peptide detection using LC-MS without protein reduction (ISO 20418-3:2020) • EN ISO 20705:2020 Textiles - Quantitative microscopical analysis - General principles of testing (ISO 20705:2019) • EN ISO 20706-1:2019 Textiles - Qualitative and quantitative analysis of some bast fibres (flax, hemp, ramie) and their blends - Part 1: Fibre identification using microscopy methods (ISO 20706-1:2019, Corrected version 2020-03) • EN ISO 2076:2021 Textiles - Man-made fibres - Generic names (ISO 2076:2021)	
CEN/TC 248/WG 37 "Textiles and textile products - Microplastics from textile sources" ¹⁴	• EN ISO 4484-1:2023 Textiles and textile products - Microplastics from textile sources - Part 1: Determination of material loss from fabrics during washing (ISO 4484-1:2023) • EN ISO 4484-3:2023 Textiles and textile products - Microplastics from textile sources - Part 3: Measurement of collected material mass released from textile end products by domestic washing method (ISO 4484-3:2023)	Currently none. A proposal for a new work item is under discussion in the WG.
CEN/TC 248/WG 39 "Textiles and textile products - Circular Economy for textile products and the textile chain" ¹⁵	• EN ISO 5157:2023 Textiles - Environmental aspects - Vocabulary (ISO 5157:2023)	• 00248762 Textiles — Circular economy for textile products — General principles and guidance • 00248761 Textiles — Circular economy for textile products — Categorisation of and requirements on non-virgin input materials

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:2722717&cs=1F93583CCBB3FAC964CFA8FE1709B3ECE

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:2922255&cs=1B1FF5082447D96C38A6CCBB8798C861D

Working group	Published standards	Work programme
		00248763 Textiles — Circular economy for textile products — Design for circularity 00248776 Textiles – Circular economy for textile products – Minimum requirements for clothing 00248786 Textiles – Circular economy for textile products – Minimum requirements for workwear 00248785 Textiles – Circular economy for textile products – Minimum requirements for bed, bath kitchen and table textiles

CEN WS "TRICK - Guidelines on data collection from Textile supply chains for the Digital Product Passport"

Submitted by the TRICK project; for commenting in CEN TC248.

CEN/TC134 Resilient, Textile, Laminate and Modular Mechanical Locked Floor Coverings¹⁶

Standardization of definitions, requirements, classification and test methods, and development of guidance documents and reports for resilient, textile, laminate and modular mechanical locked floor coverings. The main use areas for floor coverings within the scope of CEN/TC 134 are residential (homes, apartments) and commercial, (health care, education, hospitality, public buildings, offices, retail, transportation). These areas are limited to indoor use. Excluded are screeds, raised access floors, paving, surfaces for sports areas, as well as parquet, wood veneer and bamboo floorings.

This TC has currently five WGs, 96 published standards and 21 standards under development.

Working group	Work programme (related to the topic of this document)
CEN/ TC134/ WG8 – Textile floor coverings¹⁷ <i>This WG develops standards specifically for textile floor coverings.</i>	prEN 17903 – "Definition and declaration of recycled content (organic and inorganic) in textile floor coverings". The document failed FV and it as decided to stop the work. A work item applicable to all floor coverings will be started.

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:6116&cs=1DC35683E32D0CD853B6F13D4B2C314E4

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:416362&cs=1990521695BAC166CB59DBC530026CB14

CEN/TC189 Geosynthetics¹⁸

Standardization related to geosynthetics; terminology, sampling before testing, identification and marking rules, test methods, requirements related to the intended use.

This TC has currently seven WGs, 77 published standards and 13 standards under development.

Working group	Work programme
CEN TC189 WG7 Geosynthetics sustainability and environmental topics ¹⁹	00189279 Geosynthetics - Environmental performance – product category rules 00189280 Geosynthetics - Circular Economy – requirements towards materials and products 00189281 Geosynthetics – Potential release of microplastics 00189282 Geosynthetics - Release of dangerous substances

ISO TC61 Plastics – SC14 Environmental Aspects²⁰

All standardization activities in the field of plastics relating to environmental and sustainability aspects. The focus is on, but not limited to bio-based plastics, biodegradability, environmental footprint incl. carbon footprint, resource efficiency incl. circular economy, characterization of plastics leaked into the environment incl. microplastics, waste management incl. organic, mechanical and chemical recycling.

This TC has currently five WGs, 41 published standards and 12 standards under development.²¹

Working group	Published standards	Work programme
ISO/TC 61/SC 14/WG 1 “Plastics - Environmental aspects – Terminology, classifications and general guidance”	ISO 17422:2018 Plastics — Environmental aspects — General guidelines for their inclusion in standards (with CEN)	ISO/CD 15270-1.2 Plastics — Guidelines for the recovery and recycling of plastics waste — Part 1: General principles
ISO/TC 61/SC 14/WG 2 “Plastics - Environmental aspects – Biodegradability”	- EN ISO 20200:2023 Plastics — Determination of the degree of disintegration of plastic materials under simulated composting conditions in a laboratory-scale test - ISO 5430:2023 Plastics — Plastics — Marine ecotoxicity testing scheme for soluble breakdown intermediates	- ISO/AWI 25306 Plastics — Determination of the aerobic biodegradation of plastic materials in an aqueous medium — Method by analysis of evolved carbon dioxide in a closed system - ISO/AWI 25304 Plastics — Determination of the degree of biodegradation and

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:6170&cs=18DA3530C8CD9671DB4264520D7E8D155

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https://standards.cencenelec.eu/dyn/www/f?p=205:7:0:::FSP_ORG_ID:3372370&cs=1415BB9994897F2958AD412B4CA0FE48F

²⁰ <https://www.iso.org/committee/6578018.html>

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https://standards.cencenelec.eu/dyn/www/f?p=205:22:0:::FSP_ORG_ID:FSP_LANG_ID:874780.25&cs=12C4CA7D4006888F70EDD8C10E7C2AD07

Working group	Published standards	Work programme
	from biodegradable plastic materials used in and intentionally added to the marine environment – Test methods and requirements • ISO 5424:2022 Plastics — Industrial compostable plastic drinking straws • ISO 5412:2022 Biodegradable plastic shopping bags for industrial composting • ISO 5148:2022 Plastics — Determination of specific aerobic biodegradation rate of solid plastic materials and disappearance time (DT50) under mesophilic laboratory test conditions • ISO 23832:2021 Plastics — Test methods for determination of degradation rate and disintegration degree of plastic materials exposed to marine environmental matrices under laboratory conditions • ISO 23517:2021 Plastics — Biodegradable mulch films for use in agriculture and horticulture — Requirements and test methods regarding biodegradation, ecotoxicity and control of constituents • ISO 17088:2021 Plastics — Organic recycling — Specifications for compostable plastics • EN ISO 16929:2021 Plastics — Determination of the degree of disintegration of plastic materials under defined composting conditions in a pilot-scale test • EN ISO 14852:2021 Determination of the ultimate aerobic biodegradability of plastic materials in an aqueous medium — Method by analysis of evolved carbon dioxide • EN ISO 22766:2020 Plastics — Determination of the degree of disintegration of plastic materials in marine habitats under real field conditions • EN ISO 23977-1:2020 Plastics — Determination of the aerobic biodegradation of	disintegration of plastic materials under high-solids anaerobic-digestion conditions • ISO/AWI 25303 Plastics — Determination of the degree of biodegradation and disintegration of plastic materials under wet anaerobic digestion conditions • ISO/AWI 25302 Plastics — Evaluation of the dispersibility and solubility of plastic materials under marine conditions • ISO/AWI 23292 Plastics — A method for measuring the amount of bacteria in the hydrosphere biodegradability assessment • ISO/AWI 14855-2 Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions — Method by analysis of evolved carbon dioxide — Part 2: Gravimetric measurement of carbon dioxide evolved in a laboratory-scale test • ISO/CD 18957 Plastics — Determination of the aerobic biodegradation of plastic materials exposed to seawater using accelerated conditions in laboratory • ISO/DIS 15270-5 Plastics — Guidelines for the recovery and recycling of plastics waste — Part 5: Organic recycling • ISO/DIS 16636 Plastics — Simple field test of disintegration of plastics under real marine environment • ISO/FDIS 16623 Plastics — Preparation methods of sea water and sediment for marine biodegradation evaluation of plastics

Working group	Published standards	Work programme
	plastic materials exposed to seawater — Part 1: Method by analysis of evolved carbon dioxide • EN ISO 23977-2:2020 Plastics — Determination of the aerobic biodegradation of plastic materials exposed to seawater — Part 2: Method by measuring the oxygen demand in closed respirometer • EN ISO 19679:2020 Plastics — Determination of aerobic biodegradation of non-floating plastic materials in a seawater/sediment interface — Method by analysis of evolved carbon dioxide • EN ISO 22404:2019 Plastics — Determination of the aerobic biodegradation of non-floating materials exposed to marine sediment — Method by analysis of evolved carbon dioxide • EN ISO 17556:2019 Plastics — Determination of the ultimate aerobic biodegradability of plastic materials in soil by measuring the oxygen demand in a respirometer or the amount of carbon dioxide evolved • EN ISO 14851:2019 Determination of the ultimate aerobic biodegradability of plastic materials in an aqueous medium — Method by measuring the oxygen demand in a closed respirometer • ISO 13975:2019 Plastics — Determination of the ultimate anaerobic biodegradation of plastic materials in controlled slurry digestion systems — Method by measurement of biogas production • EN ISO 14855-2:2018 Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions — Method by analysis of evolved carbon dioxide — Part 2:	

Working group	Published standards	Work programme
	Gravimetric measurement of carbon dioxide evolved in a laboratory-scale test • EN ISO 18830:2016 Plastics — Determination of aerobic biodegradation of non-floating plastic materials in a seawater/sandy sediment interface — Method by measuring the oxygen demand in closed respirometer • EN ISO 14853:2016 Plastics — Determination of the ultimate anaerobic biodegradation of plastic materials in an aqueous system — Method by measurement of biogas production • EN ISO 19679:2016 Plastics — Determination of aerobic biodegradation of non-floating plastic materials in a seawater/sediment interface — Method by analysis of evolved carbon dioxide • EN ISO 20200:2015 Plastics — Determination of the degree of disintegration of plastic materials under simulated composting conditions in a laboratory-scale test • EN ISO 15985:2014 Plastics — Determination of the ultimate anaerobic biodegradation under high-solids anaerobic-digestion conditions — Method by analysis of released biogas • EN ISO 14855-1:2012 Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions — Method by analysis of evolved carbon dioxide — Part 1: General method • EN ISO 10210:2012 Plastics — Methods for the preparation of samples for biodegradation testing of plastic materials • ISO 17088:2012 Specifications for compostable plastics	
ISO/TC 61/SC 14/WG 3 “Plastics - Environmental aspects - Biobased plastics”	• ISO 5425:2023 Specifications for use of poly(lactic acid)	• ISO/AWI 16620-5 Plastics — Biobased content — Part 5: Declaration of biobased

Working group	Published standards	Work programme
	based filament in additive manufacturing applications • ISO 22526-1:2020 Plastics — Carbon and environmental footprint of biobased plastics — Part 1: General principles (also published as EN) • ISO 22526-2:2020 Plastics — Carbon and environmental footprint of biobased plastics — Part 2: Material carbon footprint, amount (mass) of CO₂ removed from the air and incorporated into polymer molecule (also published as EN) • ISO 22526-3:2020 Plastics — Carbon and environmental footprint of biobased plastics — Part 3: Process carbon footprint, requirements and guidelines for quantification (also published as EN) • ISO 22526-4:2023 Plastics — Carbon and environmental footprint of biobased plastics — Part 4: Environmental (total) footprint (Life Cycle Assessment) • ISO 16620-1:2015 Plastics — Biobased content — Part 1: General principles • ISO 16620-2:2019 Plastics — Biobased content — Part 2: Determination of biobased carbon content • ISO 16620-3:2015 Plastics — Biobased content — Part 3: Determination of biobased synthetic polymer content • ISO 16620-4:2024 Plastics — Biobased content — Part 4: Determination of biobased mass content • ISO 16620-5:2017 Plastics — Biobased content — Part 5: Declaration of biobased carbon content, biobased synthetic polymer content and biobased mass content	carbon content, biobased synthetic polymer content and biobased mass content
ISO/TC 61/SC 14/WG 4 “Plastics - Environmental aspects - Characterization of plastics leaked into the environment (including microplastics) and quality	• ISO 24187:2023 Principles for the analysis of microplastics present in the environment • ISO/TR 21960:2020 Plastics — Environmental aspects — State of knowledge and methodologies (with CEN)	• ISO/PWI 25220 Plastics — Resin loss prevention and mitigation management practices • ISO/WD 24899 A method for the extraction of biodegradable and non-biodegradable microplastics from compost

Working group	Published standards	Work programme
control criteria of respective methods”		
ISO TC61 SC14 WG5 Plastics Environmental aspects - Mechanical and chemical recycling	- ISO 5677:2023 Testing and characterization of mechanically recycled Polypropylene (PP) and Polyethylene (PE) for intended use in different plastics processing techniques - ISO/TR 23891:2020 Plastics — Recycling and recovery — Necessity of standards - ISO 15270:2008 Plastics — Guidelines for the recovery and recycling of plastics waste	- ISO/WD 15270-2 Plastics — Guidelines for the recovery and recycling of plastics waste — Part 2: Mechanical recycling - ISO/WD 15270-3 Plastics — Guidelines for the recovery and recycling of plastics waste — Part 3: Physical recycling - ISO/CD 15270-4 Plastics — Guidelines for the recovery and recycling of plastics waste — Part 4: Chemical recycling - ISO/CD 13390 Plastics — Chemical Recycling — Gasification

CEN/TC249 Plastics²²

Standardization of 1) terminology, 2) test methods, 3) specifications, classifications and designation systems, 4) environmental aspects, 5) joining systems and techniques, of plastics, plastic-based materials, semi-finished products and products (thermoplastics, thermosets, degradable plastics, bio-based polymers, thermoplastic elastomers, composites, reinforcement products for plastics, recyclates).

Rubber is excluded. Specific end-product related items are also excluded if they are covered by the scope of an existing product TC.

This TC has currently eleven WGs, a few hundreds of published standards and 68 standards under development.

Working group	Published standards	Work programme
CEN/ TC 249 Plastics - WG 9 Bio-based and biodegradable plastics ²³	- CEN/TR 17910:2022 Biodegradable plastics - Status of standardization and new prospects. - EN 17417:2020 Determination of the ultimate biodegradation of plastics materials in an aqueous system under anoxic (denitrifying) conditions -	Currently none

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Working group	Published standards	Work programme
	Method by measurement of pressure increase • EN ISO 22403:2020 Plastics — Assessment of the intrinsic biodegradability of materials exposed to marine inocula under mesophilic aerobic laboratory conditions — Test methods and requirements • EN 17228:2019 Plastics - Bio-based polymers, plastics, and plastics products - Terminology, characteristics and communication • EN 14995:2006 Plastics - Evaluation of compostability - Test scheme and specifications • EN 14987:2006 Plastics - Evaluation of disposability in waste water treatment plants - Test scheme for final acceptance and specifications • CEN/TR 15351:2006 Plastics - Guide for vocabulary in the field of degradable and biodegradable polymers and plastic items	
CEN /TC 249/ WG 11 Plastics - Plastics recycling ²⁴ <i>CEN TC249 WG11 has a large workload under the Standardisation Request M/584 on Plastics recycling and recycled plastics.</i>	• EN 15342:2007 Plastics - Recycled Plastics - Characterization of polystyrene (PS) recyclates (under revision, see work program) • EN 15343:2007 Plastics - Recycled Plastics - Plastics recycling traceability and assessment of conformity and recycled content • EN 15344:2021 Plastics - Recycled plastics - Characterization of Polyethylene (PE) recyclates (under revision, see work program) • EN 15345:2007 Plastics - Recycled Plastics - Characterisation of Polypropylene (PP) recyclates (under revision and split into 6 parts, see work program) • EN 15346:2014 Plastics - Recycled plastics -	• prEN 15342 Plastics - Recycled plastics - Characterization of polystyrene (PS) recyclates • prEN 15344 Plastics - Recycled plastics - Characterization of Polyethylene (PE) recyclates • prEN 15345 Plastics - Recycled Plastics - Characterisation of Polypropylene (PP) recyclates • FprEN 15346 Plastics - Recycled plastics - Characterization of poly(vinyl chloride) (PVC) recyclates • prEN 15347-2 Plastics - Sorted plastics wastes - Part 2: Quality grades of sorted Polyethylene (PE) wastes and specific test methods • prEN 15347-3 Plastics - Sorted plastics wastes - Part 3: Quality grades of sorted

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Working group	Published standards	Work programme
	Characterization of poly(vinyl chloride) (PVC) recyclates (under revision, see work program) • EN 15347:2007 Plastics - Recycled Plastics - Characterisation of plastics wastes (under revision and split into 6 parts) • EN 15348:2014 Plastics - Recycled plastics - Characterization of poly(ethylene terephthalate) (PET) recyclates (revised document under publication, see work program) • CEN/TR 15353:2007 Plastics - Recycled plastics - Guidelines for the development of standards for recycled plastics • CEN/TS 16010:2020 Plastics - Recycled plastics - Sampling procedures for testing plastics waste and recyclates • CEN/TS 16011:2013 Plastics - Recycled plastics - Sample preparation • CEN/TS 16861:2015 Plastics - Recycled plastics - Determination of selected marker compounds in food grade recycled polyethylene terephthalate (PET) • EN 17410:2021 Plastics - Controlled loop recycling of PVC-U profiles from windows and doors • CEN/TS 17627:2021 Plastics - Recycled plastics - Determination of solid contaminants content • EN 15348:2024 Plastics - Recycled plastics - Characterization of poly(ethylene terephthalate) (PET) recyclates	Polypropylene (PP) wastes and specific test methods • prEN 15347-4 Plastics - Sorted plastics wastes - Part 4: Quality grades of sorted poly(ethylene terephthalate) (PET) wastes and specific test methods • prEN 15347-5 Plastics - Sorted plastics wastes - Part 5: Quality grades of sorted poly(vinyl chloride) (PVC) wastes and specific test methods • prEN 15347-6 Plastics - Sorted plastics wastes - Part 6: Quality grades of sorted polystyrene (PS) wastes and specific test methods • prEN 18064-1 Plastics - Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 1: General aspects • prEN 18064-2 Plastics - Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 2 : Polyethylene (PE) • prEN 18064-3 Plastics Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 3 : Polypropylene (PP) • prEN 18064-4 Plastics - Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 4 : Poly(ethylene terephthalate) (PET) • prEN 18064-5 Plastics - Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 5 : Poly(vinyl chloride) (PVC) • prEN 18064-6 (WI=00249A3F) Plastics - Quality recommendations and basis for specifications for application of plastic recyclates in products - Part 6 : Polystyrene (PS) • prEN 18064-7 Plastics - Quality recommendations

Working group	Published standards	Work programme
		and basis for specifications for application of plastic recyclates in products - Part 7 : Acrylonitrile- butadiene- styrene (ABS) - prEN 18065 Plastics - Recycled plastics - Classification of recycled plastics based on Data Quality Levels for use and (digital) trading - prEN 18067 Plastics - Recycled plastics - Characterization of Acrylonitrile-Butadiene-Styrene (ABS) recyclates - prCEN/TR WI=00249A5G Plastics recycling — Classification of plastic recyclates as postconsumer recyclates (PCR) and postindustrial recyclates (PIR)
CEN/TC 249/WG 24 "Plastics - Environmental aspects" ²⁵	- CEN ISO/TR 21960:2020 Plastics - Environmental aspects - State of knowledge and methodologies (ISO/TR 21960:2020) - EN 17615:2022 Plastics - Environmental Aspects - Vocabulary	Currently none
CEN/TC 249/ WG26 "Plastics - Agricultural plastic products - Design-for-recycling, use, removal, collection and recycling" ²⁶	Currently none	prEN 18109 Agricultural plastic products - Design-for-recycling, installation, use, removal, collection and preparation for recycling (under approval, under M/584)

CEN/TC261 Packaging - SC4 – Packaging and Environment ²⁷

This field covers all environment related aspects of packaging. SC4 will, among other things, deal with the following aspects: - material recovery, - energy recovery, - other types of recovery, - degradability, - symbology, - terminology, - determination of environmental impacts criteria.

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Working group	Published standards	Work programme
CEN/TC 261/SC 4/WG 2 “Packaging – Packaging and Environment - Degradability and organic recovery of packaging and packaging materials” ²⁸	• EN 13432:2000 Packaging - Requirements for packaging recoverable through composting and biodegradation - Test scheme and evaluation criteria for the final acceptance of packaging • EN 14045:2003 Packaging - Evaluation of the disintegration of packaging materials in practical oriented tests under defined composting conditions • EN 14046:2003 Packaging - Evaluation of the ultimate aerobic biodegradability of packaging materials under controlled composting conditions - Method by analysis of released carbon dioxide • EN 14047:2002 Packaging - Determination of the ultimate aerobic biodegradability of packaging materials in an aqueous medium - Method by analysis of evolved carbon dioxide • EN 14048:2002 Packaging - Determination of the ultimate aerobic biodegradability of packaging materials in an aqueous medium - Method by measuring the oxygen demand in a closed respirometer • EN 17427:2022 Packaging - Requirements and test scheme for carrier bags suitable for treatment in well-managed home composting installations • EN 17428:2023 Packaging - Determination of the degree of disintegration under simulated home composting conditions	• prEN 13432rev Packaging - Requirements for packaging recoverable through composting and biodegradation - Test scheme and evaluation criteria for the final acceptance of packaging (preliminary)
CEN/TC 261/SC 4/WG 3 “Packaging – Packaging	• CEN/TR 13688:2008 Packaging - Material recycling - Report on requirements for substances and materials to prevent a	• prCEN/TR 13504rev Packaging - Material recovery - Criteria for a minimum content of recycled material (preliminary)

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Working group	Published standards	Work programme
and Environment - Material recovery" ²⁹	sustained impediment to recycling - CR 13504:2000 Packaging - Material recovery - Criteria for a minimum content of recycled material - EN 13430:2004 Packaging - Requirements for packaging recoverable by material recycling - EN 13437:2003 Packaging and material recycling - Criteria for recycling methods - Description of recycling processes and flow chart - EN 13440:2003 Packaging - Rate of recycling - Definition and method of calculation	00261469 Packaging - Quality grades for plastic packaging for recycling and measuring recycling (preliminary) 00261470 Packaging - Material recycling - Report on requirements for substances and materials to prevent a sustained impediment to recycling (preliminary)
CEN/TC 261/SC 4/WG 6 "Packaging – Packaging and Environment – Prevention" ³⁰	(not relevant)	
CEN/TC 261/SC 4/WG 7 "Packaging – Packaging and Environment – Reuse" ³¹	(not relevant)	
CEN/TC 261/SC 4/WG 8 "Packaging – Packaging and Environment - Heavy metals and other dangerous substances" ³²	- CEN/TR 13695-2:2019 Packaging - Requirements for measuring and verifying the four heavy metals and other dangerous substances present in packaging, and their release into the environment - Part 2: Requirements for measuring and verifying dangerous substances present in packaging, and their release into the environment - CR 13695-1:2000 Packaging - Requirements for measuring and verifying the four heavy metals and other dangerous substances present in	WI=00261467 Packaging - Requirements for measuring and verifying the four heavy metals and other dangerous substances present in packaging and their release into the environment - Part 1: Requirements for measuring and verifying the four heavy metals present in packaging

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Working group	Published standards	Work programme
	packaging and their release into the environment - Part 1: Requirements for measuring and verifying the four heavy metals present in packaging	
CEN/TC 261/SC 4/WG 10 “Packaging – Packaging and Environment - Design for recycling for plastic packaging products” ³³	Currently none	The complete work program of the WG is under mandate M/584. • prEN 18120-1 Packaging - Design for recycling for plastic packaging products - Part 1: Definitions and principles for design-for-recycling of plastic packaging • prEN 18120-2 Packaging - Design for recycling for plastic packaging products - Part 2: Process and governance structure to evaluate the recyclability of plastic packaging • prEN 18120-3 Packaging - Design for recycling for plastic packaging products - Part 3: Sorting evaluation process for plastic packaging • prEN 18120-4 Packaging - Design for recycling for plastic packaging products - Part 4: Guideline and protocols for PET bottles • prEN 18120-5 Packaging - Design for recycling for plastic packaging products – Part 5 – Guideline and protocols for PET rigid packaging (except bottles) • prEN 18120-6 Packaging – Design for recycling for plastic packaging products – Part 6 – Guideline and protocols for PE and PP rigid packaging • prEN 18120-7 Packaging - Design for recycling for plastic packaging products - Part 7: Guideline and protocols for PE and PP flexible packaging • prEN 18120-8 Packaging - Design for recycling for plastic packaging products - Part 8: Guideline and protocols for PS packaging

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Working group	Published standards	Work programme
		• prEN 18120-9 Packaging - Design for recycling for plastic packaging products - Part 9: Guideline and protocols for PS packaging • prEN 18120-10 Packaging - Design for recycling for plastic packaging products - Part 10: Recyclability evaluation process for plastic packaging - protocols for PET bottles • prEN 18120- 11 Packaging – Design for recycling for plastic packaging products – Part 11 – Recyclability evaluation process for plastic packaging – protocols for PET rigid packaging (except bottles) • prEN 18120-12 Packaging – Design for recycling for plastic packaging products – Part 12 – Recyclability evaluation process for plastic packaging – protocols for PE and PP rigid packaging • prEN 18120-13 Packaging – Design for recycling for plastic packaging products – Part 13 – Recyclability evaluation process for plastic packaging – protocols for PE and PP flexible packaging • prEN 18120-14 Packaging – Design for recycling for plastic packaging products – Part 14 – Recyclability evaluation process for plastic packaging – protocols for PS and XPS packaging • prEN 18120-15 Packaging - Design for recycling for plastic packaging products - Part 15: Recyclability evaluation process for plastic packaging - protocols for EPS packaging

CEN/TC466 “Circularity and Recyclability of Fishing Gear and Aquaculture Equipment”

34

Standardization in the field of circularity and recyclability of plastic-based materials in fishing gear and aquaculture equipment. Excluded is standardization work concerning fish processing, fish packaging, fish food products and general work on materials and equipment covered by other CEN/TCs.

Working group	Under publication (November 2024)
CEN/TC 466/WG 1 “Circularity and recyclability of fishing gear and aquaculture equipment - Technical requirements of CDFG (Circular Design of Fishing Gear)”	EN 17988-3:2024 Circular design of fishing gear and aquaculture equipment - Part 3. Technical requirements *
CEN/TC 466/WG 2 “Circularity and recyclability of fishing gear and aquaculture equipment - Environmental and circular requirements for fishing gear and aquaculture equipment”	- EN 17988-4:2024 Circular design of fishing gear and aquaculture equipment - Part 4. Environmental and circularity requirements and guidelines * - EN 17988-5:2024 Circular design of fishing gear and aquaculture equipment - Part 5. Circular business models for fishing gear and aquaculture equipment * - CEN/TS 18101:2024 Circular design of fishing gear and aquaculture equipment – Terms and Definitions *
CEN/TC 466/WG 3 “Circularity and recyclability of fishing gear and aquaculture equipment - Guidance on implementing CDFG (Circular Design of Fishing Gear)”	- EN 17988-1:2024 Circular design of fishing gear and aquaculture equipment - Part 1. General requirements and guidelines * - EN 17988-2:2024 Circular design of fishing gear and aquaculture equipment - Part 2. User manual and labelling * - EN 17988-6:2024 Circular design of fishing gear and aquaculture equipment - Part 6. Digitalization of gear and components *

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02

Gap analysis on standards

Identified gaps

Despite the huge effort being made on the development of new standards adequate for bio-based materials, namely textile-based ones, as shown above, some gaps are still identified, as for example:

- Determination of recycled content, both for synthetic and natural fibres (by testing, to verify claims);
- Identification of different man-made cellulosic fibres, to allow distinction between them. For example, distinction between wood-based cellulose, plant based (banana, pineapple, hemp, nettle, etc.), biological waste, is still not possible;
- Determination of the biodegradability of textiles in different environments (e.g., sea water, fresh water, landfill, soil), there are some methods but not all environments are covered, also not always ultimate biodegradation is addressed;
- Identification and quantification of micro fibres (as different from micro plastics) released during washing, with identification at least between biodegradable and non-biodegradable ones (proposal for new part 4484-4 looks at distinguishing cotton and polyester as lab method);
- Clarification on concepts like biodegradable and bio-based polymers;
- Uniformization of LCA methodology – although there are standards for LCA of bio-based products (EN 16751 and EN16760 based on ISO 14040 and ISO 14044, ISO 22526), there is not sufficient consensus over sufficient indicators and assessment methods to ensure that all products environmental impacts are assessed through comparable methodologies;

- When defining the minimum criteria for materials and products, the specific properties of recycled and bio-based materials and other alternative materials need to be considered;
- Need a larger data base towards the criteria for circularity/ sustainability of textile materials and how the testing relates to this;
- Criteria for design for sustainability need to be defined;
- Definitions/ terminology of end-of-life material becoming end-of-waste material which then goes into recycling (also referred to as feedstock for recycling); how to assess the quality of this material;

Identified gaps already being (partially) addressed

Methods are being developed or improved to cover some of the above identified gaps on standards for renewable textile materials.

For example, CITEVE, the Portuguese Textile Technological Centre for the Textile and Clothing Industries, is improving and developing new methods under the national funded project be@t -bioeconomy at textiles, as are examples:

- Determination of the degradability of textiles. This method is being developed based on ISO 11721 (Determination of resistance of cellulose-containing textiles to micro-organisms) and ISO 20200 (Determination of the degree of disintegration of plastic materials under simulated composting conditions in a laboratory-scale test), and includes the analysis of the ecotoxicity of the soil after biodegradation;
- Identification and quantification of micro fibres released during washing. This work is based on the improvement of ISO4484 (Microplastics from textile sources, Part 1: Determination of material loss from fabrics during washing, Part 2: Qualitative and quantitative analysis of microplastics), namely in the amount/concentration of fibres collected for analysis (Part 1), and the methods for the identification of the fibres (Part 2). Several techniques are being explored, such as Raman, FTIR, DSC, TGA, and pyrolysis;
- Identification of the type of cellulosic fibre. Through DNA sequencing and FTIR, work is being done to develop a method to distinguish between different natural cellulosic fibres, such as cotton, pineapple, banana, nettle, or hemp;
- Quantification of hazardous chemical substances in new fibres. Different techniques, such as chromatographic and mass spectrometric ones are being applied to identify certain substances, such as THC and CBD in hemp fibres;
- Determination of the recycled PES content. Different techniques are being applied (DSC, oscillatory plate rheology, MFI, FTIR, TGA, GPC, methanolysis and swelling of PET) in an attempt to distinguish between virgin PES and recycled PES fibres in textiles;
- Cytotoxicity of textiles. Based on ISO 10993-5 (Biological evaluation of medical devices, Part 5: Tests for in vitro cytotoxicity), a method is being developed for the determination of the in vitro cytotoxicity of textiles with different compositions.

Note: release of waste (water) during production is not included here as this is regulated by legislation.

03

Overview of certification schemes and labels

Next to standards there are also private certification schemes and labels for biobased and renewably sourced textiles and textile materials. The list below includes currently identified relevant certification schemes and labels. It is not meant to be an exhaustive list; member projects of ECOSYSTEM like STAR4BBS are working on developing a more extensive list and data base³⁵.

General information

Topic	Link
Informative website	https://www.biobasedconsultancy.com/en/about-biobased/certification-and-ecolabels
EU legislation	https://environment.ec.europa.eu/topics/plastics/biobased-biodegradable-and-compostable-plastics_en

³⁵ <https://star4bbs.eu/>

ISO 14024 type I labels

Logo	Name	Area	Further information
	EU-Ecolabel ³⁶	Europe	https://environment.ec.europa.eu/topics/circular-economy/green-claims_en
	Blue Angel – German eco label ³⁷	Germany	
	Nordic Swan Ecolabel ³⁸	Nordic Countries (Denmark, Finland, Iceland, Norway and Sweden)	The Nordic Swan Ecolabel is a: – Type I ecolabel according to the standard ISO 14024. – Trustworthy, independent, third-party certification providing companies with documentation on environmental excellence. – Member of GEN – the Global Ecolabelling Network for Type I ecolabels.

Sustainable feedstocks

Logo	Name	Further information
	Forrest stewardship council (FSC) ³⁹	
	Programme for the Endorsement of Forest Certification (PEFC) ⁴⁰	• leading global alliance of national forest certification systems. • international non-profit, non-governmental organization • promote sustainable forest management through independent third-party certification
	ISCC ⁴¹	ISCC's Objectives • Implementation of zero-deforestation • Protection of land with high biodiversity value and high carbon stock • Protection of soil, water and air • Compliance with human, labour and land rights • Measurement and reduction of greenhouse gas emissions • Traceability throughout supply chains

³⁶ https://environment.ec.europa.eu/topics/circular-economy/eu-ecolabel_en

³⁷ <https://www.blauer-engel.de/en>

³⁸ <https://www.nordic-swan-ecolabel.org/official-nordic-ecolabel/>

³⁹ <https://fsc.org/en>

⁴⁰ <https://www.pefc.org/>

⁴¹ <https://www.iscc-system.org/>

Logo	Name	Further information
	REDcert ⁴²	REDcert offers certification schemes for sustainable biomass, biofuels and bioliquids (REDcert-EU) as well as sustainable agricultural raw materials for use in the food/feed industry and biomass for material purposes respectively material purposes in the chemical industry (REDcert ²).
	Roundtable on Sustainable Biomaterials (RSB) ⁴³	
	Better Biomass ⁴⁴	Better Biomass is an international certification system for solid, liquid and gaseous biomass. The Better Biomass certificate is used to demonstrate the sustainability of the biomass used for energy, fuels or bio-based products. The underlying sustainability criteria have been established in a multi-stakeholder consensus process between private companies, government and civil society organizations and are published in the standard NTA 8080. These criteria cover important values for nature, environment and society (People, Planet, Profit).
	Roundtable on Sustainable Palm Oil (RSPO) ⁴⁵	The emphasis is on "sustainable." Sustainable palm oil has been farmed, processed, distributed, and sold responsibly with strict rules that protect animals, the environment and people who live and work in oil palm producing countries. It has involved: • Halting deforestation; • Treating communities and workers fairly; and • Protecting wildlife and the environment. RSPO's aim is to make palm oil sustainable. We work across supply chain sectors, bringing together its many stakeholders to develop sustainable practices and help make palm oil a force for good.
	Global sustainability platform for sugarcane - Bonsucro	Collectively accelerating the sustainable production and uses of sugarcane, focusing on climate action, human rights and value in the supply chain. We work across all sugarcane products and derivatives – sugar, ethanol, molasses, and bagasse in traditional and newer market sectors, from sugar and alcohol to biofuels and bioplastics.
	Round table on responsible soy (RTRS) ⁴⁶	Non-profit organisation promoting the growth of production, trade, and use of responsible soy. It works through cooperation with those in, and related to, the soy value chain, from production to consumption. It does this through:

⁴² <https://www.redcert.org/en/>

⁴³ <https://rsb.org/>

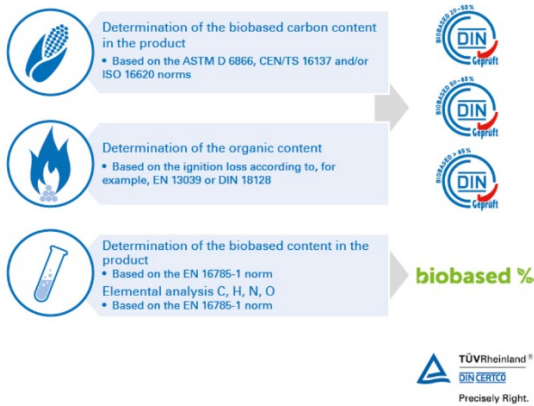
⁴⁴ <https://betterbiomass.nl/en/>

⁴⁵ <https://rspo.org/>

⁴⁶ <https://responsiblesoy.org/?lang=en>

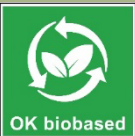
Logo	Name	Further information
		- A global platform for multi-stakeholder dialogue on responsible soy. - The development, implementation and verification of a global certification standard. RTRS' mission is to promote the growth on production, trade, and use of responsible soy through cooperation with actors in and relevant to the soy value chain from production to consumption in an open dialogue with stakeholders including producers, suppliers, manufacturers, retailers, financial institutions, civil society organizations and other relevant actors.

Bio-based content

Logo	Name	Further information
	DIN-CERTCO ⁴⁷	All products that have an organic content of at least 20 percent - with the exception of medical, toxic and fuel products - can be certified with the DIN-Geprüft logo. You have the possibility to have the complete product or a part of your product unit certified, thus minimizing the testing effort and making it more economical. The test is based on the following standards: - ASTM D 6866 – „Standard Test Methods for Determining the Biobased Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis“ - ISO 16620 – „Plastics - Biobased content“ - For the “bio-based” logo according to DIN EN 16785-1, the percentage of bio-based carbon can be indicated with a corresponding logo. - DIN EN 16785-1 - Biobased products - Biobased content - Part 1: Determination of biobased content using radiocarbon and elemental analysis (German version EN 16785-1: 2015) TESTING SPECIFICATION FOR BIOBASED PRODUCTS  The diagram illustrates the testing process for biobased products. It consists of three main steps, each with a corresponding icon and description: Determination of the biobased carbon content in the product: Based on the ASTM D 6866, CEN/TS 16137 and/or ISO 16620 norms. This step is represented by a corn icon. Determination of the organic content: Based on the ignition loss according to, for example, EN 13039 or DIN 18128. This step is represented by a flame icon. Determination of the biobased content in the product: Based on the EN 16785-1 norm, Elemental analysis C, H, N, O, and based on the EN 16785-1 norm. This step is represented by a test tube icon. The final result is a biobased % logo, which is shown alongside the DIN-CERTCO logo. The DIN-CERTCO logo is a circular emblem with the text 'DIN-CERTCO' and 'Geprüft'. TÜVRheinland® DIN-CERTCO Precisely Right.
biobased %	Biobased (NEN) ⁴⁸	The European bio-based content certification scheme is used to specify and validate the amount of biomass in a bio-based product, based on the European standard EN 16785-1:2015. This European standard provides a method of determining the bio-based content of solid, liquid and gaseous products using the radiocarbon analysis and elemental analyses.

⁴⁷ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/packaging/biobased-products/>

⁴⁸ <https://biobasedcontent.eu/en/certification/>

Logo	Name	Further information
 	TÜV Austria OK biobased ⁴⁹	On a basis of the determined percentage of renewable raw materials (% Bio-based), your product can be certified as one-star-bio-based, two-star-bio-based, three-star-bio-based or four-star-bio-based.

Compostable

Logo	Name	Further information
 	TÜV Austria⁵⁰ OK compost INDUSTRIAL (EN 13432) OK compost HOME	OK compost HOME is not based on a standard but is the basis for several standards. It seems important to remember that the OK compost HOME certification programme does not explicitly refer to a specific standard but details all the technical requirements that a product must meet in order to obtain the certification. It should also be remembered that, as a pioneer in this field, the requirements of the OK compost HOME programme, defined in 2003 and never questioned since then, have served as the basis for the drafting of several standards such as: • Australia: AS 5810 (2010) – Biodegradable plastics - Biodegradable plastics suitable for home composting • France: NF T 51800 (2015) – Plastics - Specifications for plastics suitable for home composting • Europe: EN 17427 (2022) - Packaging — Requirements and test scheme for carrier bags suitable for treatment in well-managed home composting installations
	DIN CERTCO Industrial composting ⁵¹	
	European bioplastics SEEDLING ⁵²	SEEDLING: compliance with EN 13432
 	Compostable materials Certification Scheme CMCS – UK	

⁴⁹ <https://www.tuv-at.be/okcert/certifications/ok-biobased/>

⁵⁰ <https://www.tuv-at.be/okcert/certifications/ok-compost-seedling-real/>

⁵¹ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/environmental-field/industrial-compostable-products/>

⁵² <https://www.european-bioplastics.org/>

Biodegradable

Logo	Name	Further information
  	TÜV Austria⁵³ OK biodegradable SOIL OK biodegradable WATER OK biodegradable MARINE	
   	DIN Certco⁵⁴ Soil⁵⁵ Marine⁵⁶ Additives According to DIN EN 13432⁵⁷	DIN CERTCO offers certification for mulch films, intermediates and materials according to DIN EN 17033. We test and certify according to ISO 22403, the first international standard for the assessment of intrinsic biodegradability of materials in marine environments. In addition to biodegradability, the certification scheme "Biodegradable in Marine Environments" also defines criteria for marine toxicity and disintegration. The content of heavy metals and other substances of concern are also taken into account. In the certification process, we distinguish between different types of additives: • Water- and/or solvent-based printing inks • Inorganic pigments • Organic dyes • (Color) master batches based on biodegradable materials • Biodegradable additives The testing process takes into account the following tests, depending on the standard and the nature and composition of the additives: • Chemical characterization • Test for complete biodegradability in laboratory tests • Plant compatibility • Earthworm toxicity test • Nitrification test • IR transmission spectrum The central standard for the certification of your biodegradable additives is DIN EN 13432. Depending on

⁵³ <https://www.tuv-at.be/okcert/certifications/ok-biodegradable/>

⁵⁴ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/biologische-abbaubarkeit/biodegradability/>

⁵⁵ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/environmental-field/biodegradable-in-soil/>

⁵⁶ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/environmental-field/biodegradable-in-marine-environment/>

⁵⁷ <https://www.dincertco.de/din-certco/en/main-navigation/products-and-services/certification-of-products/environmental-field/additives-according-to-din-en-13432/>

Logo	Name	Further information
		your individual requirements, we also test and certify according to one of the following standards: • ASTM D 6400 "Standard Specification of Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities" • ASTM D 6868 "Standard Specification for Labeling of End Items that Incorporate Plastics and Polymers as Coatings or Additives with Paper and Other Substrates Designed to be Aerobically Composted in Municipal or Industrial Facilities" • DIN EN 14995 "Plastics - Evaluation of Compostability - Test Scheme and Specifications" • ISO 17088 "Specification for Compostable Plastics" • ISO 18606 "Packaging and Environment - Organic recycling" • AS 4736 "Biodegradable Plastics - Biodegradable Plastics suitable for Composting and other Microbial Treatment" • AS 5810 "Biodegradable plastics — Biodegradable plastics suitable for home composting" • NF T 51-800 "Plastics - Specifications for plastics suitable for home composting" • DIN EN 17033 "Plastics - Biodegradable mulch films for use in agriculture and horticulture - Requirements and test methods" • ISO 23517 "Plastics - Soil biodegradable materials for mulch films for use in agriculture and horticulture - Requirements and test methods regarding biodegradation, ecotoxicity and control of constituents" In any case, you can prove the safety of your products for the composting process with our "DIN Tested Additives EN 13432" mark.

Chemical safety

Logo	Name	Further information
	OEKO-TEX® standard 100 ⁵⁸	This is a label for textiles tested for harmful substances. It sets the benchmark for textile safety, through all the value-chain.
	ZDHC Roadmap to zero ⁵⁹	This is a programme that aims to implement a sustainable chemical management in the textile industry. Products with the ZDHC Approved Status comply with the ZDHC MRSL.

⁵⁸ <https://www.oeko-tex.com/en/our-standards/oeko-tex-standard-100>

⁵⁹ <https://www.roadmaptozero.com/>

Recycled or organic content

Logo	Name	Further information
	GOTS / OCS	Both oriented to products containing content from biologic production (through rastreability schemes), GOTS also considers chemical substances, processes and other management criteria.
	BETTER COTTON ⁶⁰	Mass balance systems
	GRS / RCS	Both oriented to products with recycled content (through rastreability schemes), GRS also considers chemical substances, and social and environmental principals.

More sustainable practices

Logo	Name	Further information
	Bluesign ^{®61}	This label ensures that all textiles and products have met the strictest standards throughout the supply chain in relation to chemical consumption, carbon emission, water consumption, energy consumption, worker health and safety.
	OEKO-TEX [®] MADE IN GREEN ⁶²	Textile and leather products with this label are produced more sustainably in socially responsible workplaces.
	The Green Button ⁶³	Government-run certification label for sustainable textiles. «It is the first certification label that systematically evaluates whether companies take responsibility for respecting human rights and environmental standards in their supply chains.»
	Textile exchange (meta label) ⁶⁴	Focus on the sourcing of fibres and raw materials through farming and extraction (from earth, plants and animals). Materials covered are: • Plant Fibers (Cotton, Hemp) • Manmade Cellulosics (Viscose) • Synthetics (Biosynthetics, Nylon, Polyester) • Animal Fibers & Materials (Down, Leather, Wool) Textile exchange has the following standards: • Global Recycled Standard • Organic Content Standard • Recycled Claim Standard • Responsible Wool Standard • Content Claim Standard • Responsible Alpaca Standard

⁶⁰ <https://bettercotton.org/pt/>

⁶¹ <https://www.bluesign.com/en/>

⁶² <https://www.oeko-tex.com/en/our-standards/oeko-tex-made-in-green>

⁶³ <https://gruener-knopf.de/en>

⁶⁴ <https://textileexchange.org/>

Logo	Name	Further information
		• Responsible Down Standard • Responsible Mohair Standard There are currently ongoing efforts to merge these standards into one Materials Matter Standard.

Further reading

<https://telaketju.turkuamk.fi/uploads/2020/12/e9e1655f-sertifikaatit-selvitys.pdf> (in Finnish)

Siegelklarheit : <https://www.siegelklarheit.de/en/>

ECOSYSTEM

European Community of Practice for a Sustainable Textile Ecosystem

Overview of standardisation committees, published and under development standards, and gap analysis for renewable materials/ textiles

Website: www.ecosystem.eu · LinkedIn: ECOSYSTEM

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